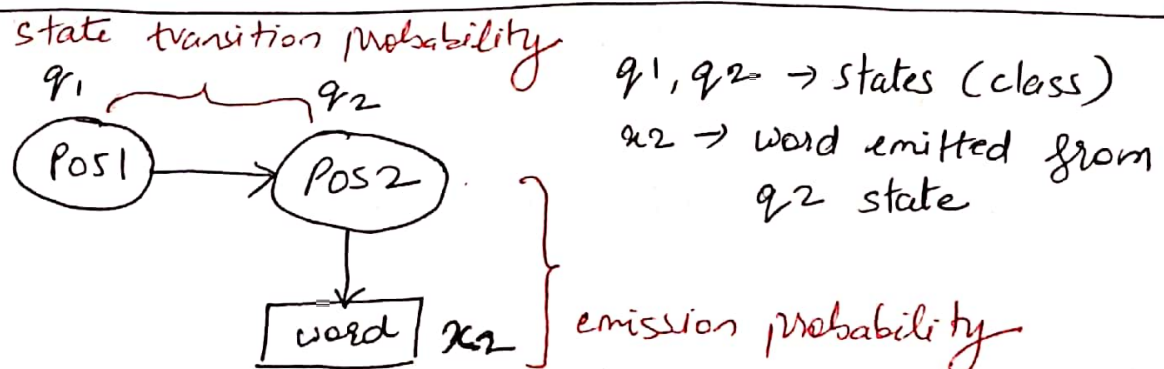


POS Tagging Problem

- There are 2 types of approaches
 - 1) generate the data from the class
 - 2) ~~the~~ class generates data
- Consider data available in the form (d, c)
 - where $d \rightarrow$ observations (words)
 - $c \rightarrow$ hidden classes (Pos tags)
- There are 2 types of models
 - 1) Generative model - eg: HMM
 - 2) Discriminative model - eg: maximum Entropy (maxEnt) model, Conditional Random Field

Hidden Markov model (HMM) - Generative Model



$$P(\text{POS2} | \text{word}, \text{POS1}) = \underbrace{P(\text{word} | \text{POS2})}_{\substack{\downarrow \\ \text{emission} \\ \text{probability} \\ \text{(using Bayes rule)}}} \times \underbrace{P(\text{POS2} | \text{POS1})}_{\substack{\downarrow \\ \text{state transition} \\ \text{probability} \\ \text{(using bigram)}}}$$

Emission Probability matrix calculation

(2)

- 1) From given corpus, identify unique words (tokens).
- 2) Identify total part of speech tags present in corpus.
- 3) ~~Identify~~ Assign correct Part of Speech (pos) tag for every token
- 4) List tokens as rows and pos tags as columns (vice-a-versa also works)
- 5) ~~For~~ From this data, Bayes rule emission probability is computed.
- 6) For 'N' tokens and 'm' pos tags, size of emission matrix will be $N \times M$.

State Transition Probability matrix calculation

- 1) Use pos tags found from emission probability matrix
- 2) For 'm' pos tags, create matrix of size $m \times m$
- 3) Compute the matrix values using bigram concept
- 4) This provides data for state transition probability

Viterbi Decoding Algorithm

(3)

Consider the sentence:

"Assured to back the silver coin"

~~Data~~ marking all the possible pos tags for each unique word

word possible pos tags	w ₁ assured	w ₂ to	w ₃ back	w ₄ the	w ₅ silver	w ₆ coin
	VBD: Past Tense	To: TD	RB: Adverb	NH: Noun	NN: Noun	NN: Noun
	VBN: Participle		NN: Noun	Deter- miner	JJ: Adjective	VB: Verb
	JJ: Adjective		JJ: Adjective		VB: Verb	
			VB: Verb			
total	3	1	4	2	3	2

∴ Total number of ways =

$$3 \times 1 \times 4 \times 2 \times 3 \times 2 = 144$$

= 144 possibilities

↑

too complex and tedious

Thus to reduce the number of possibilities

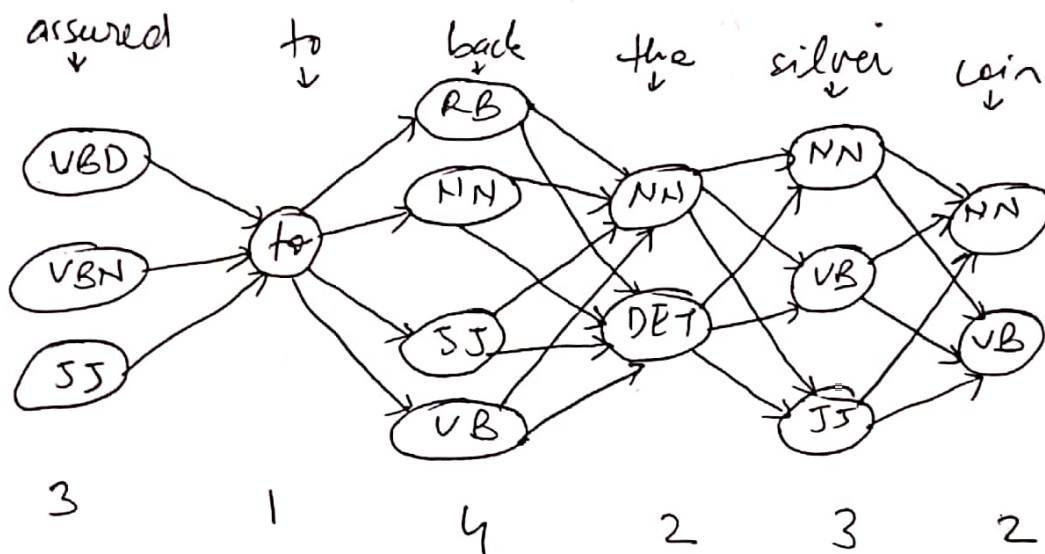
Viterbi Decoding Algorithm is used

Viterbi Decoding Algorithm

(4)

- It is based on dynamic programming
- It saves the result for further purposes
i.e. it saves the POS tag having highest probability value

Working of Viterbi Decoding Algorithm



Total no. of possibilities without Viterbi = $3 \times 1 \times 4 \times 2 \times 3 \times 2$
= 144

Applying Viterbi Algorithm

eg ① consider word "assured"
possible POS tags = VBD, VBN, JJ

② Compute $P(VBD)$, $P(VBN)$, $P(JJ)$

③ Save highest probability value from step 2.
so let's consider $P(VBD)$ is highest

④ repeat ~~set~~ steps 1-3 for next word "to"

⑤ compute final probability for "to" POS tag

as $P(\text{to-POS tag}) \times P(\text{assured-VBD})$
i.e. $P(\text{current word-POS tag}) = P(\text{current word-POS}) \times P(\text{previous word-POS})$

Example:

(5)

Q) Consider the given corpus data / ~~training~~ training corpus data:

<s> martin justin can watch will <e>

<s> spot will watch martin <e>

<s> will justin spot martin <e>

<s> martin will pat spot <e>

<s> → start of sentence

<e> → end of sentence

Find pos tag and remove pos ambiguities for the following test data

Test Data:

<s> justin will spot will <e>

Consider following pos tags:

N: Noun, m: Modal Verb, V: Verb

step 1: Assign correct pos tag to training corpus / label training corpus with pos tag

<s>	martin	gustin	can	watch	will	<e>
	N	N	M	V	N	
<s>	spot	will	watch	martin	<e>	
	N	M	V	N		
<s>	will	gustin	spot	martin	<e>	
	M	N	V	N		
<s>	martin	will	pat	spot	<e>	
	N	M	V	N		

N: noun, m: modal verb, V: verb

step 2: calculate Emission Probability matrix

$$P(\text{word}|\text{class}) = \frac{P(\text{class, word})}{P(\text{class})}$$

unique words (tokens)	Pos tags	P(class)		
		N	M	V
martin		4/9	0	0
gustin		2/9	0	0
will		<u>1/9</u>	3/4	0
spot		2/9	0	1/4
can		0	1/4	0
watch		0	0	2/4
pat		0	0	1/4
Total (POS tags)		9	4	4

Example calculation: $P(\text{will}|\text{N})$

$$= \frac{P(\text{will, N})}{P(\text{N})}$$

$$= \frac{\text{count}(\text{will, N})}{\text{count}(\text{N})} = \underline{\underline{\frac{1}{9}}}$$

step 3: calculate state transition matrix / transition matrix using bigram probability

$$P(W_i | W_{i-1}) = \frac{\text{count}(W_{i-1}, W_i)}{\text{count}(W_{i-1})}$$

(W _{i-1}) Pos tags	(W _i) Pos tags	N	M	V	<E>
<s>		3/4	1/4	0	0
N		1/9	<u>3/9</u> <u>1/3</u>	1/9	4/9
M		1/4	0	<u>3/4</u>	0
V		1	0	0	0

Example calculation:

$$P(V|M) = \frac{\text{count}(M, V)}{\text{count}(M)}$$

$$= \underline{\underline{\frac{3}{4}}}$$

Step 4: Assign Pos tags to test data

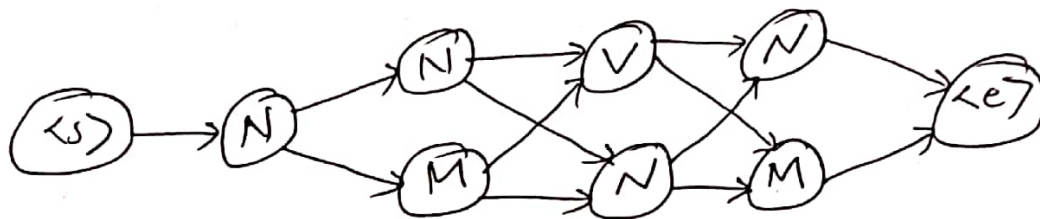
(7)

Consider Test Data

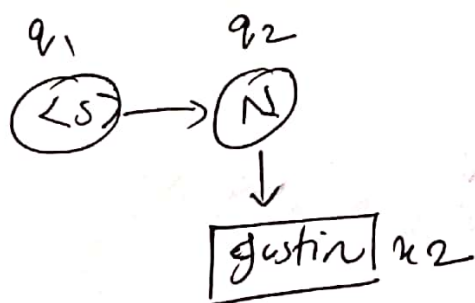
<s> gustin will spot will <e>

	gustin	will	spot	will
	N	N	V	N
		M	N	M
Total	1	2	2	2

$\therefore$ Total possibilities = $1 \times 2 \times 2 \times 2 = 8$



Step 5: calculate the probability of pos tag for every word



For word: gustin as noun

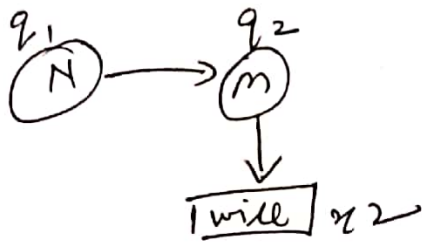
$$\begin{aligned}
 &P(N | \text{gustin}, \langle s \rangle) \\
 &= P(\text{gustin} | N) \times P(N | \langle s \rangle) \\
 &\quad \text{emission matrix} \quad \text{transition matrix} \\
 &= \frac{2}{9} \times \frac{3}{4} = \frac{1}{6}
 \end{aligned}$$

$$\therefore P(N | \text{gustin}, \langle s \rangle) = \frac{1}{6}$$

word: will

(2)

will as modal verb

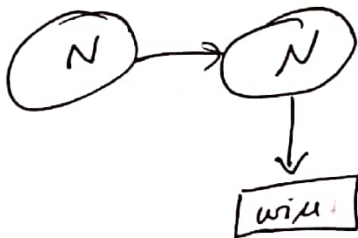


$$P(m|will, N) = P(will|m) \times P(m|N) \\ = \frac{3}{4} \times \frac{1}{3} = \frac{1}{4}$$

multiply with previous word probability

$$\therefore P(m|will, N) = \frac{1}{4} \times \frac{1}{6} = \frac{1}{24}$$

will as Noun



$$P(N|will, N) = P(will|N) \times P(N|N) \\ = \frac{1}{9} \times \frac{1}{9} = \frac{1}{81}$$

multiply with previous word probability

$$\therefore P(N|will, N) = \frac{1}{81} \times \frac{1}{6} = \frac{1}{486}$$

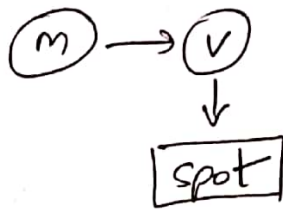
$$\therefore \frac{1}{24} > \frac{1}{486}$$

$$\therefore P(m|will, N) > P(N|will, N)$$

$$\therefore \boxed{will \rightarrow \text{modal verb}}$$

$$\therefore \boxed{P(m|will, N) = \frac{1}{24}}$$

spot as Verb



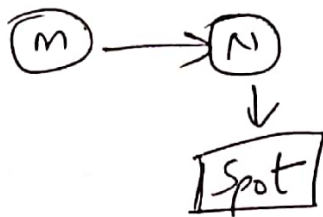
word: spot

(9)

$$P(V|spot, m) = P(spot|V) \times P(V|m) \\ = \frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$$

$$\therefore P(V|spot, m) = \frac{3}{16} \times \frac{1}{24} = \frac{1}{128}$$

spot as Noun



$$P(N|spot, m) = P(spot|N) \times P(N|m) \\ = \frac{2}{9} \times \frac{1}{4} = \frac{1}{18}$$

$$\therefore P(N|spot, m) = \frac{1}{18} \times \frac{1}{24} = \frac{1}{432}$$

$$\therefore \frac{1}{128} > \frac{1}{432}$$

$$\therefore P(V|spot, m) > P(N|spot, m)$$

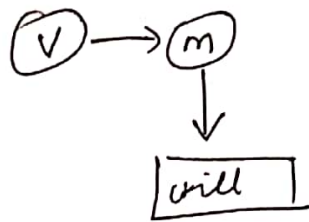
$$\therefore \boxed{\text{spot} \rightarrow (V) \text{ Verb}}$$

$$\boxed{P(V|spot, m) = \frac{1}{128}}$$

word: will

(10)

will as modal verb



$$P(M|will, V) = P(will|M) \times P(M|V)$$
$$= \frac{3}{4} \times 0$$

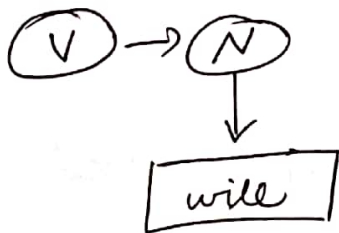
$$= \frac{3}{4} \times \frac{1}{1000}$$

simple smoothing

$$= \frac{3}{4000}$$

$$P(M|will, V) = \frac{3}{4000} \times \frac{1}{128} = \frac{1}{512000}$$

will as Noun



$$P(N|will, V) = P(will|N) \times P(N|V)$$
$$= \frac{1}{9} \times \frac{1}{1} = \frac{1}{9}$$

$$P(N|will, V) = \frac{1}{9} \times \frac{1}{128} = \frac{1}{1152}$$

$$\therefore \frac{1}{1152} > \frac{1}{512000}$$

$$\therefore P(N|will, V) > P(M|will, V)$$

$$\therefore \boxed{will \rightarrow (N) \text{ Noun}}$$

$$\boxed{P(N|will, V) = \frac{1}{1152}}$$

Step 6 : Result of POS Tagging

(11)

<s> Justin will spot will <e>

N M V N